Predictive Lab Mid exam -1 9-09-2026

Marks : 25 marks (10 for each problem and 5 for viva)

1. Using the Medical Cost Personal Dataset (insurance.csv), create a Power BI Python Visual to predict medical insurance charges using a Linear Regression model. Should contain the following (Marks also depend on how well you design and present the visualizations.)
   (a) Create cards for: Total Medical Charges , Average Medical Charges , Number of Patients
   (b) Create a calculated column called BMI Category using DAX and classify patients as: Underweight , Normal , Overweight , Obese
   (c) Graph between independent and dependent variable.
   (d) Plot an Actual vs. Predicted Medical Charges visualization using the Power BI Python Visual.
   (e) Show Predict values for last three rows in the data set in the graph.

2. Using the Titanic dataset (train.csv), preprocess the data and convert the categorical target variable Survival_Status (Yes/No) into numerical values using Label Encoding. Build a Decision Tree Classifier to classify passengers into two classes—Survived and Did Not Survive and implement both pre-pruning and post-pruning using python code. Compare the unpruned, pre-pruned, and post-pruned models using accuracy, precision, recall, F1-score, and confusion matrix.